OpenFGA performs well in relationship-rich applications but needs to be combined with other systems to manage contextual decisions.

8.5.4 AWS IAM

AWS IAM remains the most widely used authorization system globally. It is deeply integrated into the AWS ecosystem and capable of fine-grained, context-aware control, but it is limited to AWS.

- Principal: IAM users, roles, or federated identities; attributes through tags and principal tags.
- Action: AWS API operations (such as `"s3:GetObject"`); service-specific.
- Resource: AWS objects identified by ARNs; supports tagging and resource-based policies.
- Context: Extensive through global and service-specific condition keys; tag-aware.
- Strengths: Comprehensive AWS integration, fine-grained, globally scalable, mature tooling.
- Limitations: AWS-specific; not applicable outside AWS.

IAM delivers the most mature, large-scale authorization in the industry, but only inside AWS.

8.5.5 Cedar

Cedar draws lessons from all of these: declarative policies like XACML, embedding and developer-focused tools like OPA, dynamic relationship modeling like OpenFGA, and tight integration like IAM, while keeping portability and a clear type system.

- Principal: Typed principals with schema-validated attributes.
- Action: Typed actions with schema enforcement and explicit policy scoping.
- Resource: Typed resources with schema-validated attributes.